

HaaV

Highly amphibious / autonomous Vehicle

Team: UW–Madison

College of Engineering

We are a student team building an uncrewed autonomous boat from the ground up.



40 mph

Speed

Under 80 lbs

Weight

Realtime LTE

With LoRa backup

Ground up means simulation + modeling first

We are building the model from first principles, mixing empirical and rule-of-thumb decisions with dynamic programming. Each constraint cuts the space of possible boats, and our model finds the fastest one based on current variables. As we build and test, we keep updating the model and adding constraints: a hybrid of simulation and empirical data.

Engineers from across UW–Madison

Lead members have interned at Tesla, Northrop Grumman, Xcel Energy, Fincantieri and Milwaukee Tool.

Industrial

Optimization, composites and manufacturing

Mechanical

Structures, CAD, fabrication

Electrical

Power, battery and high voltage

Computer

Firmware, software, controls

Our primary phases

Hull

Plane fast and stay light.

Hybrid planing composite hull
Dimensions shaped by weight
Optimization shaped by hydrodynamics

Electric

Every watt-hour to speed.

High-voltage battery storage
Propulsion sized by the model
Propulsion and cooling co-designed

Autonomous

Manage drift and controls.

GPS, IMU and LIDAR sensing
Fundamental feedback protocols
Software plans the route

Timeline

Oct 25

Hull design frozen

Nov 1

Preliminary design review

Nov 10

Critical and long-lead parts are ordered

Nov 20

Mechanical and electrical design frozen

Nov 20–Dec 20

Build the hull, composite parts and any custom-milled parts

Jan 8

First float and tow test

Jan 8–30

Jan 8–30

Jan 31

Feb 1–14

Feb 15

Until April

Manufacture and assemble everything

Build likely spare parts, in parallel

First nautical mile

Slow drives on the water

Start increasing speed

Testing / Tuning

Want to get involved?

We are looking for funds, parts, advisors and supporters.
Email Marley or head to our site for more information.

bit.ly/haav

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